Finite-Sample Reconstruction:
Rates, Witnesses, and the Honest Bridge

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Abstract

We study finite-sample reconstruction in measure-preserving systems. Given a time series $x_1, \ldots, x_n$ from a system $(X, \mathcal{B}, \mu, T)$ and a single observable h, we ask whether reconstruction — the condition that the delay algebra $\sigma(h, h \circ T, \ldots)$ generates $\mathcal{B}$ modulo μ — is detectable from data alone, at what rate, and with what certificates.

We prove three theorems. The *Algebra Theorem* shows that the empirical σ-algebra approximation error $\hat{\delta}(L, n)$ concentrates around the true error $\delta(L)$, and that the elbow stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d-dimensional state space, without oracle knowledge of the mixing rate, the lag, or the smoothness index. The *Dynamics Theorem* shows that under uniform separation, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz and estimated delay vectors certify point separation at rate $n^{-\beta/(2\beta+d)}$. The *Conjunction Theorem* shows that for deterministic T, algebra separation and metric separation are the same event — not merely correlated, but identical — and that both witnesses certify this from data under reconstruction and uniform separation.

The key structural insight is a conditional variance identity expressing $\delta(L)$ as an integral over unseparated trajectory pairs. This identity yields a third computable witness: the collision entropy $H_2(\nu_L) := -\log(\mu \otimes \mu)(R_L)$ of the delay-vector distribution, which diverges if and only if $\delta(L) \to 0$ under a fibre mixing condition. The three witnesses — algebraic, geometric, and information-theoretic — all measure the same failure of separation across delay fibres.

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1 Introduction

1.1 Context and question

This paper is the fourth in a series on observable dynamics. Paper I [6] constructs the canonical probability measure from an observable algebra via Carathéodory and Stone extension. Paper II [5] establishes Koopman–Perron duality and the predictive semigroup. Paper III [7] proves the reconstruction theorem: the observable algebra $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ generates $\mathcal{B}$ modulo μ if and only if the observable $h : X \to \mathbb{R}$ is a measure-theoretic embedding.

Paper IV asks: *what does the reconstruction theorem look like from finite data?*

Given a time series $x_1, \ldots, x_n$ drawn from a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and a single observable h, can one determine from data alone whether reconstruction holds? If so, at what rate? And with what certificates?

We answer these questions with three theorems: the Algebra Theorem, the Dynamics Theorem, and the Conjunction Theorem. The three theorems are calibrated: each is proved under exactly the hypotheses its proof uses, and the scope of each is named precisely.

1.2 Three honest theorems

The Algebra Theorem (Section 5). Define the σ-algebra approximation error

$$\delta(L) = \sup_{S\in\mathcal{B}} \inf_{E\in\mathcal{O}_h^{(L)}} \mu(S\triangle E),$$

which measures how much of $\mathcal{B}$ is not yet captured by the finite-lag algebra $\mathcal{O}_h^{(L)} = \sigma(h, h\circ T, \ldots, h\circ T^L)$. This quantity is estimable from data as $\hat{\delta}(L,n)$ (defined in Section 2). The elbow stopping rule $\hat{L}^*$ fires when $\hat{\delta}(L,n)$ stops decreasing. Under exponential mixing with resolvability constant $C_\lambda > 4$, $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d-dimensional state space, without any oracle knowledge of the mixing rate, the lag $L^*(n)$, or the smoothness index s. This is the main finite-sample rate result.

The Dynamics Theorem (Section 6). The delay map $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$ encodes the trajectory of x under the observable. When $T \in C^r$ and the uniform separation condition (US) holds (infimum of the smallest singular value of $D\Phi_h^{(L)}$ over X is bounded away from zero), $\Phi_h^{(L)}$ is bi-Lipschitz. The empirical delay vectors then constitute a second witness: they certify that points are being separated by the dynamics. The Dynamics Theorem shows that this kernel witness converges to the true separation at rate $n^{-\beta/(2\beta+d)}$ and names the places where it can fail.

The Conjunction Theorem (Section 7). The two witnesses measure the same event. For deterministic T, the TV separation $d_L(x,x') = 1$ if and only if x and x' are separated by $\mathcal{O}_h^{(L)}$ — not merely correlated, but identical. This algebraic identity ($\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$) is the Markov bridge. Under reconstruction and (US), both witnesses fire correctly and certify the same object. The theorem also names two failure modes: reconstruction failure makes the algebra witness fail; violation of (US) makes the kernel witness fail even when reconstruction holds.

1.3 What is new

The σ-algebra approximation error $\delta(L)$ is observable without knowing the mixing rate, the dimension d, or the smoothness s. Standard adaptive methods (cross-validation, Lepski's method) patch the smoothness oracle but leave the dimension as a residual unknown. The elbow rule for $\hat{\delta}(L,n)$ patches both simultaneously by operating upstream of the rate: it asks "has the algebra captured $\mathcal{B}$?" rather than "at what rate is the approximation converging?"

This paper establishes three computable witnesses for reconstruction:

(i) $\hat{\delta}(L,n)$ — the algebraic witness (Section 5);

(ii) $\hat{d}_L(x,x')$ — the delay-map separation witness (Section 6);

(iii) $\hat{H}_2(\nu_L^{(n)})$ — the collision entropy witness (Section 5, Corollary 5.12); computationally simpler than (i): no optimisation over sets, only delay-vector histogram counts.

Witnesses (i) and (iii) are asymptotically equivalent under the fibre mixing condition (Corollary 5.12); witness (ii) provides the geometric certificate that the state space is being faithfully separated. The conditional variance identity (Lemma 3.1, Section 3) expresses the algebraic error $\delta(L)$ as an integral over unseparated pairs, identifying the precise obstruction to equivalence between (i) and (iii) as lack of fibre mixing.

The binary TV separation result (Lemma 7.1) is specific to deterministic systems. For stochastic T, the kernel $\Pi_h^{(L)}(x,\cdot)$ is not a Dirac delta, and Hölder continuity in TV is both non-trivial and necessary. This is an open direction.

1.4 Organisation

Section 2 introduces the objects used throughout, including fibre structure and collision probability. Section 3 proves the conditional variance identity, the bias bound, and the easy direction $\delta(L) \leq \frac{1}{2}(\mu \otimes \mu)(R_L)$. Section 4 proves concentration of $\hat{\delta}(L,n)$. Sections 5, 6, and 7 prove the three main theorems; Section 5 also contains the entropy characterisation of reconstruction (Corollary 5.12). Section 8 discusses open problems.

2 Setup and Notation

2.1 The measure-preserving system

Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d-dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \to \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} \;:=\; \operatorname{essinf}_X \rho \;>\; 0.$$

Let $T : X \to X$ be a μ-preserving measurable map. Paper IV makes no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

2.2 The observable and the delay map

Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \to \mathbb{R}^{L+1}, \qquad \Phi_h^{(L)}(x) \;=\; \big(h(x),\, h(Tx),\, \ldots,\, h(T^L x)\big).$$

The *observable σ-algebra at lag L* is

$$\mathcal{O}_h^{(L)} \;:=\; \sigma(\Phi_h^{(L)}) \;=\; \sigma\big\{h \circ T^k : 0 \leq k \leq L\big\}.$$

The *delay-polynomial class* of degree $D \geq 1$ is

$$\mathcal{A}_h^{(L)} \;:=\; \big\{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree} \leq D \text{ on } \mathbb{R}^{L+1}\big\} \;\subseteq\; L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

2.3 The approximation error

The central quantity of this paper is the σ-algebra approximation error introduced in Section 3:

$$\delta(L) \;:=\; \delta(\mathcal{O}_h^{(L)}) \;=\; \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E).$$

This measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. Three properties, proved in Papers I–III, are used here without proof:

(a) *Monotonicity:* $\delta(L)$ is non-increasing in L.

(b) *Reconstruction:* $\delta(L) \to 0$ if and only if $\bigcup_L \mathcal{O}_h^{(L)}$ generates $\mathcal{B}$ modulo μ-null sets.

(c) *Takens regime:* if (T, h) satisfy (G1)–(G2) below and X is a smooth manifold, then $\delta(L) = 0$ for all $L \geq L_0$ where $L_0 \leq d - 1$.

2.4 Fibre structure and collision probability

The delay map $\Phi_h^{(L)}$ partitions X into *fibres*: for each $z \in \mathbb{R}^{L+1}$, define

$$F_z := (\Phi_h^{(L)})^{-1}(z), \qquad p_z := \mu(F_z), \qquad \mu_z := \mu(\cdot \mid F_z).$$

Call z an *ε-large fibre* if $p_z \geq \varepsilon$. The push-forward $\nu_L := (\Phi_h^{(L)})_*\mu$ is the distribution of delay vectors.

The set of unseparated pairs $R_L := \{(x, x') : \Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')\}$ decomposes as $R_L = \bigsqcup_z F_z \times F_z$, giving

$$(\mu \otimes \mu)(R_L) = \int p_z^2 \, d\nu_L(z). \tag{1}$$

This quantity is the *collision probability* of the delay-vector distribution: it is the probability that two independent draws from μ produce the same delay vector.

2.5 The data model

We observe a time series $x_1, x_2, \ldots, x_n$ which is an orbit segment of T: $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ. The empirical measure is $\mu_n := \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$.

Remark 2.1 (What we do not assume)**.** We do not assume the orbit is dense, that T is ergodic, or that the time average equals the space average. The convergence results in this paper hold under the mixing hypothesis stated explicitly in each result; ergodicity is neither assumed globally nor used silently.

2.6 Structural hypotheses

The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

- (G1) *No short periodic orbits:* T has no periodic orbit of period $\leq L$ in $\mathrm{supp}(\mu)$. This ensures the orbit points $x, Tx, \ldots, T^L x$ are distinct μ-a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.
- (G2) *Non-degenerate observable:* $h \in C^2(X)$ with $dh \neq 0$ μ-a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.
- (US) *Uniform separation:* $\inf_{x \in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ-a.e. Condition (US) is used only in the Dynamics and Conjunction theorems (Sections 6 and 7).

Remark 2.2 (Gap between (G1)–(G2) and (US))**.** Hypotheses (G1)–(G2) yield $D\Phi_h^{(L)}$ full rank μ-a.e. (a set of full measure), but allow the rank to drop on a μ-null set. (US) requires uniform positivity of $\sigma_{\min}$ across all of X, including the μ-null exceptional set. The gap is genuine and cannot be closed without additional hypotheses.

2.7 The empirical witness

The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) := \sup_{p \in \mathcal{P}_D^{(L)},\, \|p\|_{L^2(\mu)} \leq 1} \left\| p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] \right\|_{L^2(\mu_n)}^2,$$

where $\mathcal{P}_D^{(L)}$ is the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 4.6).

The main result of Section 4 is that $\hat{\delta}(L,n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C\,n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

3 The Bias Bound and the Algebra Side

3.1 The bias bound

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub-σ-algebra. Define the *σ-algebra approximation error*

$$\delta(\mathfrak{m}) \;:=\; \sup_{S\in\mathcal{B}} \inf_{E\in\mathfrak{m}} \mu(S \triangle E). \tag{2}$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ-null sets.

Lemma 3.1 (Conditional variance identity)**.** *For any $S \in \mathcal{B}$:*

$$\mathbb{E}\big[\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})\big] \;=\; \frac{1}{2}\int_{R_L} \big|\mathbb{1}_S(x) - \mathbb{1}_S(x')\big|^2 \, d(\mu\otimes\mu)(x,x').$$

Proof. By disintegration, $(\mu\otimes\mu)|_{R_L} = \int \mu_z \otimes \mu_z \, d\nu_L(z)$. On each fibre F_z:

$$\tfrac{1}{2}\int_{F_z\times F_z} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 \, d(\mu_z\otimes\mu_z) \;=\; \mu_z(S)\,\mu_z(S^c).$$

Integrating with weight p_z: $\frac{1}{2}\int_{R_L} |\mathbb{1}_S - \mathbb{1}_{S'}|^2 \, d(\mu\otimes\mu) = \int p_z\, \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. Since $\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})(x) = \mu_z(S)\mu_z(S^c)$ on F_z, both sides equal $\int \mu_z(S)\mu_z(S^c)\, d\nu_L(z)$. A standalone development appears in [4]. □

Corollary 3.2 (Easy direction)**.** $\delta(L) \;\le\; \frac{1}{2}\,(\mu\otimes\mu)(R_L)$.

Proof. In Lemma 3.1, bound $|\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 \;\le\; 1$ and take the supremum over S using $\delta(L) = \sup_S \mathbb{E}[\mathrm{Var}(\mathbb{1}_S \mid \mathcal{O}_h^{(L)})]$. □

Lemma 3.3 (Bias bound)**.** *For every $f \in L^\infty(\mu)$,*

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S\triangle E) \le \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2-best $\mathfrak{m}$-measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \;\le\; \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 \;=\; \mu(S\triangle E) \;\le\; \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \le M$, the layer cake representation gives $f = \int_{-M}^{M} \mathbb{1}_{S_t}\, dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\,\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2,

$$f - \mathbb{E}[f \mid \mathfrak{m}] \;=\; \int_{-M}^{M} \big(\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big)\, dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2} \;\le\; \int_{-M}^{M} \big\|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big\|_{L^2}\, dt \;\le\; 2M \cdot \delta(\mathfrak{m})^{1/2}.$$ □

Remark 3.4. The constant 2 is not optimal but the exponent $\frac{1}{2}$ is sharp: taking $X = [0,1]$, $\mathfrak{m} = \{\emptyset, X\}$, $f = \mathbb{1}_{[0,1/2]}$ gives $\delta(\mathfrak{m}) = \frac{1}{2}$ and $\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} = \frac{1}{2} = \delta(\mathfrak{m})^{1/2}/\sqrt{2}$.

3.2 The three-term decomposition

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ-algebra at lag L, where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 3.5 (Three-term decomposition)**.** *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \;\leq\; \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{bias} \;+\; \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{approx.} \;+\; \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{statistical}, \quad (3)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (2) *and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.*

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 3.3. $\square$

Remark 3.6 (Scope)**.** Lemma 3.3 holds for any sub-σ-algebra $\mathfrak{m}$; no dynamical structure is needed. The dynamical hypothesis enters only through the approximation and statistical terms, which require the image $\Phi_h^{(L)}(X)$ to be a manageable geometric object.

3.3 The observable reverse lower bound

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n.

Lemma 3.7 (Observable reverse lower bound)**.** *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \;\geq\; \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 *as $n \to \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in* (2) *at lag L and $\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ is the empirical measure.*

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \;\leq\; \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 \;=\; \mu(S^* \triangle E) \;\geq\; \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \to 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2-projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

Remark 3.8 (Closed loop)**.** Lemmas 3.5 and 3.7 together bound $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)}$ from both sides. The upper bound (Lemma 3.5) says the error is at most $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C \cdot n^{-s/(2s+d)}$. The lower bound (Lemma 3.7) says it is at least $\frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$. The detectability threshold is the n at which the statistical floor $C' n^{-s/(2s+d)}$ drops below $\frac{\delta(L)^{1/2}}{2}$: reconstruction becomes detectable at sample size $n \gtrsim \delta(L)^{-(2s+d)/s}$.

4 Concentration of the Empirical Witness

This section proves that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$. The argument has four steps: uniform boundedness of the function class (Lemma 4.1), McDiarmid concentration around the mean (Proposition 4.3), a Glivenko–Cantelli bound for the LLN gap (Lemma 4.5), and the estimator bias via the Nadaraya–Watson kernel regression (Proposition 4.7). The full convergence is then Theorem 4.9.

Throughout this section the hypotheses of Proposition 5.3 hold: (T, h) satisfies (G1)–(G2), X is a compact smooth d-manifold, μ has smooth positive density, and $d > 2s$.

4.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \le 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 4.1 (Uniform boundedness)**.** *Define*

$$M_{D,L} \;:=\; \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \operatorname{essinf}_X \rho > 0$ *is the infimum of the density of* μ*. Then for every* $p \in \mathcal{P}_D^{(L)}$*:*

$$\|p\|_{L^\infty} \;\le\; M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to the unit box by setting $y_i = x_i/a$; polynomials of degree $\le D$ in $x \in [-a, a]^{L+1}$ correspond to polynomials $\tilde{p}$ of degree $\le D$ in $y \in [-1, 1]^{L+1}$. Expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \le D}$, where $P_\alpha(y) = \prod_{i=1}^{L+1} P_{\alpha_i}(y_i)$ and $\|P_k\|_{L^\infty[-1,1]} = 1$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives

$$|\tilde{p}(y)| \;\le\; \sum_{|\alpha| \le D} |c_\alpha| \;\le\; N(D, L+1)^{1/2} \Big(\sum_{|\alpha| \le D} |c_\alpha|^2 \Big)^{1/2},$$

where $N(D, L+1) = \binom{D+L+1}{L+1}$ counts the basis elements. From the L^2 orthogonality $\|P_\alpha\|^2_{L^2[-1,1]} = \prod_i (2\alpha_i + 1)^{-1}$, the coefficient norm satisfies

$$\sum_{|\alpha| \le D} |c_\alpha|^2 \;\le\; (2D+1)\, \|\tilde{p}\|^2_{L^2},$$

since the maximum of $\prod_i (2\alpha_i + 1)$ over $|\alpha| \le D$ is $2D + 1$, achieved at $\alpha = (D, 0, \ldots, 0)$ — the constraint $|\alpha| \le D$ forces all weight onto one coordinate, giving an L-independent maximum. Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\mathrm{vol})}$ and $\|p\|_{L^2(\mathrm{vol})} \le \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. ☐

Remark 4.2. The L-dependence of $M_{D,L}$ is entirely in $N(D, L+1)^{1/2} \sim L^{D/2}/\sqrt{D!}$ for $L \gg D$. The factor $(2D+1)^{1/2}$ depends only on degree. For fixed $D = \lceil s \rceil$ and $\|h\|_{L^\infty} < 1$ (so $2a < 1$), $M_{D,L}$ decays exponentially in L; for $\|h\|_{L^\infty} \ge \frac{1}{2}$ (equivalently $2a \ge 1$), $M_{D,L}$ grows polynomially in L. The subsequent arguments require $\|h\|_{L^\infty} \ge \frac{1}{2}$.

4.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} \;=\; \Big\{x \mapsto \big(p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x)\big)^2 \;:\; p \in \mathcal{P}_D^{(L)},\; \|p\|_{L^2(\mu)} \le 1\Big\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 4.6). Then $\hat{\delta}(L,n) = \sup_{f\in\mathcal{F}_{L,D}} \frac{1}{n}\sum_{i=1}^n f(x_i)$.

Proposition 4.3 (Concentration, $\star$)**.** *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \ge 1$:*

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \;\le\; 2\exp\Big(-\frac{c\,n\,t^2}{M_{D,L}^4 \cdot D^d}\Big). \tag{$\star$}$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C'\,n^{-2s/(2s+d)}$:

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > \varepsilon_n/2\big) \;\le\; 2\exp\Big(-c'\,n^{(d-2s)/(2s+d)}\Big) \;\to\; 0.$$

Proof. Step 1 (Boundedness). By Lemma 4.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \le 1$ satisfies $\|p\|_{L^\infty} \le M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p|\Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \le 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x_i' changes $\hat{\delta}(L,n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality:

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \;\le\; 2\exp\Big(-\frac{2n^2t^2}{n\cdot(B_{L,D}^2/n)^2}\Big) \;=\; 2\exp\Big(-\frac{n\,t^2}{2B_{L,D}^4/n}\Big).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [2]:

$$\mathbb{E}[\hat{\delta}(L,n)] - \delta(L) \;\le\; 2\,\mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma, since $f = (p - \cdot)^2$ is $2B_{L,D}$-Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \;\le\; 2B_{L,D}\,\mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d-dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [3]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \le C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \le C\,M_{D,L}\sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P\big(|\hat{\delta}(L,n) - \delta(L)| > t\big) \;\le\; 2\exp\Big(-\frac{c\,n\,t^2}{M_{D,L}^4\,D^d}\Big).$$

Setting $t = \varepsilon_n/2 = C'n^{-2s/(2s+d)}/2$, the exponent becomes

$$c\,n\,\varepsilon_n^2/(M_{D,L}^4\,D^d) \;\propto\; n^{1-4s/(2s+d)} \;=\; n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. □

Remark 4.4 (The $d > 2s$ hypothesis)**.** The condition $d > 2s$ is necessary for the concentration to be useful at the rate ε_n. It is not an artefact: when $d \le 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [1] is technically available but left to future work; Paper IV takes $d > 2s$ as a standing hypothesis.

4.3 Source 1: the LLN gap

Lemma 4.5 (LLN gap)**.** *With notation as above:*

$$\sup_{p\in\mathcal{P}_D^{(L)}}\Big|\|p-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu_n)}-\|p-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu)}\Big| \;\le\; C\sqrt{D^d/n}$$

almost surely as $n\to\infty$, where C depends on $(d,D,M_{D,L})$.

Proof. The class $\{(p-\mathbb{E}[p|\Phi_h^{(L)}])^2 : p\in\mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [3] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. □

4.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 4.6 (Nadaraya–Watson estimator)**.** For bandwidth $h_n>0$ and kernel $K:\mathbb{R}^{L+1}\to\mathbb{R}_{\ge 0}$ (compactly supported, symmetric, $\int K=1$):

$$\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}(x)] \;:=\; \frac{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)\,p(x_j)}{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)}.$$

Proposition 4.7 (Source 2 estimator bias)**.** *Assume:*

(i) $T,h\in C^r$ with $r\ge 2$, μ smooth positive density ρ with $\rho_{\min}>0$.

(ii) $\Phi_h^{(L)}$ has full rank μ-a.e. (follows from (G1)–(G2) and Lemma 5.2).

(iii) T is exponentially mixing with rate $\lambda>0$ and $\|h\|_{L^\infty}\ge\frac{1}{2}$.

(iv) The NW bandwidth satisfies $h_n\sim n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ with $d_{\mathrm{eff}}(L)=\min(L+1,d)$ and $\beta\ge 1$.

Then there exists $C_2(D,L)>0$ such that for all $p\in\mathcal{P}_D^{(L)}$:

$$\mathbb{E}\|\hat{\mathbb{E}}[p\mid\Phi_h^{(L)}]-\mathbb{E}[p\mid\Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\le\; C_2(D,L)\cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))},$$

uniformly in $p\in\mathcal{P}_D^{(L)}$, where $C_2(D,L)=2M_{D,L}^2C_1^2A^{2\beta}$ and A is a constant depending on $(\beta,d_{\mathrm{eff}}(L),\rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu=\int\mu_z\,d((\Phi_h^{(L)})_*\mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z=\{\Phi_h^{(L)}=z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z-\mu_{z'}\|_{\mathrm{TV}} \;\le\; C_1\,|z-z'|^\beta,\qquad \beta=1\ \text{(Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r},\|h\|_{C^r},\rho_{\min})$. (The optimal exponent $\beta=r-1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta=1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 4.1: for $p\in\mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)}\le 1$,

$$\big|\mathbb{E}[p\mid\Phi_h^{(L)}=z]-\mathbb{E}[p\mid\Phi_h^{(L)}=z']\big| \;=\; \Big|\int p\,d\mu_z-\int p\,d\mu_{z'}\Big| \;\le\; M_{D,L}\cdot C_1\cdot|z-z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p \mid \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L} C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [3] for a Hölder(β) regression function on a $d_{\mathrm{eff}}(L)$-dimensional domain gives:

$$\mathbb{E}\|\hat{\mathbb{E}}[p|\Phi_h^{(L)}] - \mathbb{E}[p|\Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\le\; \Lambda^2_{D,L}\, h_n^{2\beta} + \frac{C_K\, M^2_{D,L}}{n\, h_n^{d_{\mathrm{eff}}(L)}\, f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_*\mu$. The $M^2_{D,L}$ factor appears in both terms. Balancing: $\Lambda^2_{D,L} h_n^{2\beta} = C_K M^2_{D,L}/(n h_n^{d_{\mathrm{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta + d_{\mathrm{eff}}(L))}$ where A depends only on $(\beta, d_{\mathrm{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L, or p. The $M^2_{D,L}$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^*, both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta + d_{\mathrm{eff}}(L))}$ with $C_2(D, L) = 2M^2_{D,L} C_1^2 A^{2\beta}$. Since the bound is p-free, it holds uniformly over $\mathcal{P}_D^{(L)}$. □

Remark 4.8 (Scope)**.** Proposition 4.7 is proved under exponential mixing with $\|h\|_{L^\infty} \ge \frac{1}{2}$. The polynomial mixing case and the regime $\|h\|_{L^\infty} < \frac{1}{2}$ require a different estimator and are left open.

4.5 Full convergence

Theorem 4.9 (Full convergence $\hat{\delta}(L, n) \to \delta(L)$)**.** *Under the hypotheses of Proposition 4.7 and with $d > 2s$:*

$$\mathbb{E}\big|\hat{\delta}(L, n) - \delta(L)\big| \;\le\; C\sqrt{D^d/n} \;+\; C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta + d_{\mathrm{eff}}(L))}$$

and $P(|\hat{\delta}(L, n) - \delta(L)| > \varepsilon_n) \to 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L, n) - \delta(L)| \;\le\; \underbrace{|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]|}_{\text{concentration, } (\star)} \;+\; \underbrace{|\mathbb{E}[\hat{\delta}(L, n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C\sqrt{D^d/n}$ (Lemma 4.5); Source 2 is bounded by $C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta + d_{\mathrm{eff}}(L))}$ (Proposition 4.7). The concentration term is bounded by $(\star)$ (Proposition 4.3). Combining at $t = \varepsilon_n/2$ gives the probability statement. □

5 The Algebra Theorem

This section assembles the results of Sections 3 and 4 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T.

5.1 Effective dimension and the piecewise rate

Definition 5.1 (Effective dimension)**.** The *effective dimension at lag L* is

$$d_{\mathrm{eff}}(L) \;:=\; \mathrm{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x \in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ-a.e.)

Lemma 5.2 (d_{eff} step function)**.** *Under (G1)–(G2) with X a compact smooth d-manifold and μ smooth positive density:*

$$d_{\mathrm{eff}}(L) \;=\; \min(L + 1,\, d) \quad \textit{for all } L \ge 0.$$

Proof. The upper bound $d_{\text{eff}}(L) \le \min(L+1, d)$ holds because $D\Phi_h^{(L)}|_x$ is a linear map from T_xX (dimension d) to $\mathbb{R}^{L+1}$.

For the lower bound we use the Sard–Smale theorem (Smale [8] framework) applied to two failure maps.

Case $L+1 \le d$. Rank $< L+1$ at x means the cotangent vectors $\omega_j(x) := (DT^j|_x)^*(dh|_{T^jx})$ are linearly dependent. Define the failure map

$$\Psi : C^2(X) \times X \times \mathbb{P}^L \to T^*X, \quad (h, x, [\lambda]) \;\mapsto\; d\Big(\sum_j \lambda_j \, h \circ T^j\Big)\Big|_x.$$

By (G1), the orbit points T^jx are distinct μ-a.e., so bump functions at T^jx can prescribe any cotangent vector at x independently. Hence $D_h\Psi$ is surjective onto T_x^*X (dimension d), and $\Psi^{-1}(0)$ is a Banach submanifold of codimension d. The projection to $C^2(X)$ is Fredholm of index $d + L - d = L$, so by Sard–Smale the regular values form a residual set $R \subset C^2(X)$. For $h \in R$, the failure locus in X has dimension $\le L < d$, hence μ-measure zero.

By (G2), $dh \neq 0$ μ-a.e., which forces every critical point of $F_\lambda = \sum_j \lambda_j h \circ T^j$ to lie in the codimension-d submanifold $S_\lambda = \{dh_x \in \operatorname{span}\{\omega_1(x), \dots, \omega_L(x)\}\}$ (a finite set for each $[\lambda]$, since X is d-dimensional). Hence h is a regular value of π, i.e., $h \in R$.

Case $L+1 > d$. Immersion failure requires $\exists\, v \neq 0$ with $D\Phi_h^{(L)}|_x \cdot v = 0$. Define

$$\Xi : C^2(X) \times (TX \setminus 0) \to \mathbb{R}^{L+1}, \quad (h, (x, v)) \;\mapsto\; \big(dh_{T^jx}(DT^j|_x \cdot v)\big)_{j=0}^{L}.$$

By (G1), $D_h\Xi$ is surjective onto $\mathbb{R}^{L+1}$, so $\Xi^{-1}(0)$ has codimension $L+1$ and the projection to $C^2(X)$ is Fredholm of index $2d - 1 - (L+1) = 2d - L - 2$. For $L \ge d$: $2d - L - 2 \le d - 2 < d - 1$, so the failure locus in X has dimension $< d - 1$ and μ-measure zero. Condition (G2) places h in the regular-value set by the same argument as Case 1. $\square$

Proposition 5.3 (Piecewise pre-Takens rate). *Assume (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f \in \text{Hölder}(s) \cap L^\infty(\mu)$, $d > 2s$, and $T \in C^r$ with $r \ge 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D = D^*(n, L)$ chosen to balance approximation and statistical terms. Then for each $L \ge 0$:*

$$\big\|f - \hat{f}_n^{(L)}\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} \;+\; C(f, d, s) \cdot n^{-s/(2s + d_{\text{eff}}(L))}$$

with probability tending to 1, where $d_{\text{eff}}(L) = \min(L+1, d)$. In the Takens regime ($L \ge d-1$): $d_{\text{eff}}(L) = d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 3.5). The bias term is bounded by $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}$ (Lemma 3.3). By Lemma 5.2, $\Phi_h^{(L)}(X)$ is a $d_{\text{eff}}(L)$-dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f \mid \mathcal{O}_h^{(L)}](x) = g^*(\Phi_h^{(L)}(x))$ for some $g^* \in \text{Hölder}(s)$; standard polynomial approximation on a $d_{\text{eff}}(L)$-dimensional smooth domain gives approximation error $\le C_{\text{approx}} \cdot D^{-s/d_{\text{eff}}(L)}$. The statistical term is $\le C_{\text{stat}}(D^{d_{\text{eff}}(L)}/n)^{1/2}$ by Proposition 4.3 with d replaced by $d_{\text{eff}}(L)$. Balancing: $D^* \sim n^{d_{\text{eff}}(L)/(2s+d_{\text{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\text{eff}}(L))})$. $\square$

5.2 The oracle lag and the elbow stopping rule

Definition 5.4 (Oracle lag)**.** The *oracle lag* is

$$L^*(n) \;:=\; \min\big\{L \ge 0 : \delta(L) \le C_0 \cdot n^{-2s/(2s+d)}\big\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

Remark 5.5 (Post-saturation self-consistency)**.** Under exponential mixing $\delta(L) \leq Ae^{-\lambda L}$, balancing $e^{-\lambda L/2} \sim n^{-s/(2s+d)}$ gives $L^*(n) \sim \frac{2s}{\lambda(2s+d)} \log n$. For large n, $L^*(n) \gg d-1$, so the oracle lag lies in the Takens regime and the rate $n^{-s/(2s+d)}$ is self-consistent. Under polynomial mixing $\delta(L) \leq AL^{-\alpha}$, $L^*(n) \sim n^{2s/(\alpha(2s+d))} \to \infty$, also post-saturation.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L, n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 5.6 (Elbow stopping rule)**.** Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* \;:=\; \min\bigl\{L \geq 0 : \hat{\delta}(L+1, n) - \hat{\delta}(L, n) > -\tau_n\bigr\}.$$

Theorem 5.7 (Elbow location)**.** *Assume the hypotheses of Proposition 5.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1 - e^{-\lambda})Ae^{\lambda}/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then*

$$P\bigl(\hat{L}^* = L^*(n)\bigr) \;\to\; 1 \quad \textit{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n) + 1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 4.9, $|\hat{\delta}(L, n) - \delta(L)| \leq \varepsilon_n = C' n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^* + 1, n) - \hat{\delta}(L^*, n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L+1) \geq \Delta(L) \geq C_\lambda \cdot C' n^{-2s/(2s+d)}$. Tracking via Theorem 4.9:

$$\hat{\delta}(L, n) - \hat{\delta}(L+1, n) \;\geq\; C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n \;=\; (C_\lambda - 2)C' n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n/C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C' n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 5.8 (Scope of the elbow theorem)**.** The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n / L^*(n) \to 0$ relative to ε_n, so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here.

□

5.3 The Algebra Theorem

Theorem 5.9 (Algebra Theorem)**.** *Assume:*

(i) Hypotheses of Proposition 5.3: (G1)–(G2), X compact smooth d-manifold, μ smooth positive density, $f \in$ Hölder$(s) \cap L^\infty(\mu)$, $d > 2s$, $T \in C^r$, $r \geq 2$.

(ii) Exponential mixing: $\delta(L) \leq Ae^{-\lambda L}$ for some $A, \lambda > 0$.

(iii) Resolvability: $C_\lambda > 4$.

(iv) Exponential mixing for the estimator bias (Proposition 4.7) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

Let $\hat{L}^*$ be the elbow stopping rule (Definition 5.6) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then

$$\big\|f - \hat{f}_n^{(\hat{L}^*)}\big\|_{L^2(\mu)} \;=\; O_p\big(n^{-s/(2s+d)}\big).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for Hölder(s) *functions on a d-dimensional domain. No oracle input (mixing rate, $L^*(n)$, Sobolev index s) is required.*

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 5.7:

$$\begin{aligned}
\big\|f - \hat{f}_n^{(\hat{L}^*)}\big\|_{L^2(\mu)} &= \big\|f - \hat{f}_n^{(L^*(n))}\big\|_{L^2(\mu)} \\
&\le 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&\le 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f,d,s) \cdot n^{-s/(2s+d)} \\
&= O(n^{-s/(2s+d)}).
\end{aligned}$$

□

Remark 5.10 (What is and is not required)**.** The proof uses: (a) Lemma 5.2 to lock the rate at $n^{-s/(2s+d)}$ once $L \ge d-1$; (b) Theorem 4.9 to track $\delta(L)$ from data; (c) Theorem 5.7 to locate $L^*(n)$ without knowing the mixing rate; (d) Proposition 5.3 to bound the error at $L^*(n)$. The conditions $C_\lambda > 4$ and $\|h\|_{L^\infty} \ge \frac{1}{2}$ are load-bearing: removing them makes either the elbow unresolvable or the bias estimate (Proposition 4.7) unavailable. Polynomial mixing is not covered.

5.4 Entropy characterisation of reconstruction

Definition 5.11 (Fibre mixing)**.** Fix $\varepsilon > 0$. An event $S^* \in \mathcal{B}$ is (c,ε)*-mixing across fibres* if there exists $c > 0$ such that

$$\mu_z(S^*)\, \mu_z\big((S^*)^c\big) \;\ge\; c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

This condition is implied by ergodicity of T and is independent of (US). It is not a standing assumption of Paper IV; it is used only in Corollary 5.12 below.

The collision probability (1) defines a natural information-theoretic quantity:

$$H_2(\nu_L) \;:=\; -\log\big((\mu \otimes \mu)(R_L)\big) \;=\; -\log \int p_z^2 \, d\nu_L(z). \tag{4}$$

This is the *Rényi-*2 *(collision) entropy* of the delay-vector distribution.

Corollary 5.12 (Entropy characterisation of reconstruction)**.** *Under the fibre mixing condition of Definition 5.11:*

$$\delta(L) \to 0 \quad\Longleftrightarrow\quad H_2(\nu_L) \to +\infty.$$

Equivalently, reconstruction occurs if and only if the probability that two independently drawn delay vectors coincide vanishes as $L \to \infty$. Thus reconstruction is equivalent to the vanishing of collision probability $(\mu \otimes \mu)(R_L)$ introduced in Section 2.4.

Proof. Corollary 3.2 gives $\delta(L) \le \frac{1}{2}(\mu \otimes \mu)(R_L)$. For the reverse, the (c,ε)-mixing condition on a $\delta(L)$-near-maximizer S^* gives $(\mu \otimes \mu)(R_L) \le \frac{1}{c}\,\delta(L) + C(c)\,\varepsilon$ (proved in [4], Theorem 3.3), so $\delta(L) \to 0 \iff (\mu \otimes \mu)(R_L) \to 0$. Since $(\mu \otimes \mu)(R_L) = e^{-H_2(\nu_L)}$, this is equivalent to $H_2(\nu_L) \to +\infty$. □

Remark 5.13 (Empirical entropy witness)**.** The empirical analogue of $H_2(\nu_L)$ is

$$\hat{H}_2\Big(\nu_L^{(n)}\Big) \;:=\; -\log\left(\frac{1}{n^2} \sum_{i,j=1}^{n} \mathbb{1}\big[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)\big]\right),$$

the negative log of the empirical collision probability. This is the third computable witness for reconstruction. It is *computationally simpler* than $\hat{\delta}(L, n)$: it requires no optimisation over sets, only histogram counts over delay vectors. Under fibre mixing, Corollary 5.12 shows that $\hat{L}^*$ could equivalently be defined as the first L at which $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n. A detailed treatment of $\hat{H}_2$ as a stopping rule appears in [4], Section 4.

6 The Dynamics Theorem

The Algebra Theorem (Section 5) does not require any regularity of the predictive kernel $\Pi_h^{(L)}$. This section proves that, when $T \in C^r$, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz under uniform separation (US), and the estimated delay vectors certify point separation.

6.1 Lipschitz continuity of the delay map

Lemma 6.1 (Upper Lipschitz bound)**.** *Assume $T \in \mathrm{Diff}^r(X)$, $h \in C^r(X)$, $r \geq 1$. Then $\Phi_h^{(L)}$ is Lipschitz:*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\leq\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X,$$

where $C_L := \|Dh\|_{L^\infty} \cdot \big(\sum_{j=0}^{L} \|DT^j\|_{L^\infty}^2\big)^{1/2}$.

Proof. Component j of $\Phi_h^{(L)}$ is $h \circ T^j$. By the chain rule and compactness of X: $|h(T^j x) - h(T^j x')| \leq \|Dh\|_{L^\infty} \cdot \|DT^j\|_{L^\infty} \cdot d_X(x, x')$. Taking Euclidean norm over $j = 0, \ldots, L$ gives the result. □

Lemma 6.2 (Local lower Lipschitz bound)**.** *Under (G1)–(G2) with $L \geq d-1$ (Takens regime), $T \in C^r$, $r \geq 2$: for μ-a.e. $x \in X$, there exists a neighbourhood U_x and a constant $c_x = \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ such that*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\geq\; c_x \cdot d_X(x, x') \quad \forall\, x' \in U_x.$$

Proof. By Lemma 5.2, $D\Phi_h^{(L)}|_x$ has full rank d for μ-a.e. x when $L \geq d-1$, so $\sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ μ-a.e. The lower bound on U_x follows from the inverse function theorem applied to the immersion $\Phi_h^{(L)}$ at x. □

Remark 6.3 (Gap between local and global lower bound)**.** Lemma 6.2 gives a *local* lower bound $c_x > 0$ varying over X. (G1)–(G2) guarantee $\sigma_{\min} > 0$ *μ-a.e.*, but not on all of X. Condition (US) imposes $c := \inf_X \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, which is strictly stronger. The gap is genuine: there exist pairs (T, h) satisfying (G1)–(G2) but violating (US) on a μ-null set.

6.2 Uniform separation

The one-step predictive kernel for deterministic T is the Dirac family $\Pi_h^{(L)}(x, \cdot\,) = \delta_{\Phi_h^{(L)}(x)}$. The TV separation pseudometric is therefore

$$d_L(x, x') \;:=\; \|\Pi_h^{(L)}(x, \cdot\,) - \Pi_h^{(L)}(x', \cdot\,)\|_{\mathrm{TV}} \;=\; \mathbb{1}\big[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')\big] \;\in\; \{0, 1\}.$$

Separation is binary: $d_L(x, x') = 1$ if and only if x and x' are distinguished by $\mathcal{O}_h^{(L)}$.

Theorem 6.4 (Uniform separation under (US))**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, $L \geq d-1$. Then $\Phi_h^{(L)}$ is bi-Lipschitz on X:*

$$c \cdot d_X(x, x') \;\leq\; |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \;\leq\; C_L \cdot d_X(x, x') \quad \forall\, x, x' \in X.$$

For any estimator $\hat{\Gamma}_h^{(n)}$ with uniform error $|\hat{\Gamma}_h^{(n)}(x) - \Phi_h^{(L)}(x)| \le \varepsilon_n$ uniformly in x, the empirical separation $\widehat{d}_L(x,x') := |\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')|$ satisfies

$$\widehat{d}_L(x,x') \;\ge\; c \cdot d_X(x,x') - 2\varepsilon_n.$$

In particular, if $d_L(x,x') = 1$ (i.e. $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$), then $\widehat{d}_L(x,x') > 0$ for all n large enough that $\varepsilon_n < c \cdot d_X(x,x')/2$.

Proof. Upper bound: Lemma 6.1. Lower bound: (US) directly gives $|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \ge c \cdot d_X(x,x')$ for all $x, x' \in X$. For the empirical bound: by the triangle inequality, $|\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')| \ge |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| - 2\varepsilon_n \ge c \cdot d_X(x,x') - 2\varepsilon_n$. If $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ then by bi-Lipschitz $d_X(x,x') \ge |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')|/C_L > 0$, so the right side is eventually positive. □

Remark 6.5 (The delay map is directly observed)**.** In the noiseless setting (h observed exactly), the delay vectors $\Phi_h^{(L)}(x_i)$ are known to the observer directly, so $\hat{\Gamma}_h^{(n)} = \Phi_h^{(L)}$ and $\varepsilon_n = 0$. In the presence of additive observation noise, ε_n follows the NW regression rate (Proposition 4.7 with $d_{\text{eff}} = d$), so $\varepsilon_n \to 0$.

6.3 The Dynamics Theorem

Theorem 6.6 (Dynamics Theorem)**.** *Assume (G1)–(G2), (US), $T \in C^r$, $r \ge 2$, μ smooth positive density on compact d-manifold X, $L \ge d-1$. Let $\hat{\Gamma}_h^{(n)}$ be the empirical delay map estimator with uniform error $\varepsilon_n = C'' n^{-\beta/(2\beta+d)}$ (Proposition 4.7). Then:*

- *(a)* (Convergence) $\|\hat{\Gamma}_h^{(n)} - \Phi_h^{(L)}\|_{L^\infty} \to 0$ *at rate* $n^{-\beta/(2\beta+d)}$.
- *(b)* (Separation) *For $\mu \otimes \mu$-a.e. (x,x') with $x \neq x'$: $\widehat{d}_L(x,x') > 0$ for all large n.*
- *(c)* (Scope) *Condition (US) is not a consequence of (G1)–(G2). Without (US), the kernel witness $\widehat{d}_L$ can vanish on a positive-measure set of pairs even when $d_L = 1$ and reconstruction holds.*

Proof. (a) Proposition 4.7 with $d_{\text{eff}}(L) = d$ gives $\varepsilon_n \to 0$ at rate $n^{-\beta/(2\beta+d)}$.

(b) Since μ is non-atomic (smooth positive density), $\mu \otimes \mu$ almost every pair satisfies $d_X(x,x') > 0$. For such pairs $d_L(x,x') = 1$ (reconstruction and injectivity of $\Phi_h^{(L)}$), and by Theorem 6.4, $\widehat{d}_L(x,x') \ge c \cdot d_X(x,x') - 2\varepsilon_n$. For fixed $x \neq x'$, $c \cdot d_X(x,x') > 0$ and $\varepsilon_n \to 0$, so the right side is eventually positive.

(c) The proof of Theorem 6.4 uses (US) in the lower bound. Without it, there exist points x where $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ along sequences $x'_n \to x$, and $|\Phi_h^{(L)}(x'_n) - \Phi_h^{(L)}(x)|/d_X(x,x'_n) \to 0$. Any estimator inherits this degeneracy. □

7 The Conjunction Theorem

This section formalises the complete reconstruction diagnostic and proves that both witnesses certify the same event when reconstruction and (US) hold.

7.1 The honest bridge

The delay vector $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$ is a trajectory of the Markov chain on h-space with one-step kernel Π_h. The σ-algebra $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ is exactly the information generated by L steps of this Markov chain starting from x. This gives an algebraic identity:

$$\sigma\big(\Phi_h^{(L)}\big) \;=\; \mathcal{O}_h^{(L)},$$

which is not an approximation but a definition. Consequently, the σ-algebra separation event and the metric separation event are identical:

Lemma 7.1 (Algebra separation equals metric separation)**.** *For deterministic T and $\mu \otimes \mu$-a.e. (x, x'):*

$$d_L(x, x') = 1 \iff x \text{ and } x' \text{ are separated by } \mathcal{O}_h^{(L)}.$$

These are the same event, not merely correlated.

Proof. For deterministic T: $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$, so $d_L(x, x') = \|\delta_{\Phi_h^{(L)}(x)} - \delta_{\Phi_h^{(L)}(x')}\|_{\mathrm{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')]$. And x is separated from x' by $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ if and only if $\exists E \in \mathcal{O}_h^{(L)}$ with $\mathbb{1}_E(x) \neq \mathbb{1}_E(x')$, i.e., if and only if $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ (since $\mathcal{O}_h^{(L)}$ is generated by the level sets of $\Phi_h^{(L)}$). The two conditions are identical. $\square$

Under reconstruction ($\delta(L) = 0$, equivalently $\Phi_h^{(L)}$ injective μ-a.e.) and (US), injectivity holds everywhere, so $d_L(x, x') = 1$ for $\mu \otimes \mu$-a.e. $x \neq x'$.

7.2 The joint diagnostic

Definition 7.2 (Joint reconstruction diagnostic)**.** Fix tolerances $\tau_n, \eta_n > 0$. Define

$$\text{Reconstruct}(L, n) := \big\{\hat{\delta}(L, n) \leq \tau_n\big\} \wedge \big\{\widehat{d}_L(x, x') > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}\big\}.$$

Theorem 7.3 (Conjunction Theorem)**.** *Assume the hypotheses of Theorem 5.9 (for the algebra side) and Theorem 6.6 (for the dynamics side). Set $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$ and $\eta_n = \sqrt{\varepsilon_n}$ where $\varepsilon_n = C' n^{-2s/(2s+d)}$. Then:*

(a) (Conjunction fires under reconstruction + (US)). *Under $\delta(L) = 0$ for $L \geq L^*(n)$ and (US):*

$$P\big(\text{Reconstruct}(\hat{L}^*, n)\big) \to 1 \quad \textit{exponentially in } n.$$

(b) (Algebra witness fails when reconstruction fails). *If $\delta(L) > 0$ for all L: $P(\hat{\delta}(L, n) \leq \tau_n) \to 0$ for every fixed L, and $\hat{L}^* \to \infty$.*

(c) (Kernel witness fails when (US) fails). *If $\sigma_{\min}(D\Phi_h^{(L)}|_x) \to 0$ on a positive-measure set S: $P(\widehat{d}_L > \eta_n$ for $\mu \otimes \mu$-most pairs$) \to 0$ even when $\hat{\delta}(L, n) \to 0$.*

In particular: the conjunction is not achievable when either hypothesis fails.

Proof. Part (a). By Theorem 5.7, $P(\hat{L}^* = L^*(n)) \to 1$ exponentially. On $\{\hat{L}^* = L^*(n)\}$: (i) $\hat{\delta}(\hat{L}^*, n) \leq \tau_n$ whp by Theorem 4.9 and the definition of $L^*(n)$ ($\delta(L^*(n)) = 0$ under reconstruction). (ii) By Theorem 6.6(b), $\widehat{d}_{\hat{L}^*}(x, x') > 0$ for $\mu \otimes \mu$-a.e. (x, x') with $x \neq x'$, eventually. The set $\{d_X(x, x') \leq (2\varepsilon_n + \eta_n)/c\}$ has $\mu \otimes \mu$-measure $\to 0$ as $\eta_n/c \to 0$, so $\widehat{d}_{\hat{L}^*} > \eta_n$ for $\mu \otimes \mu$-most pairs.

Part (b). If $\delta(L) > 0$ for all L, then by Theorem 4.9, $\hat{\delta}(L, n) \geq \delta(L) - \varepsilon_n \geq \delta(L)/2 > \tau_n$ with high probability for all fixed L, so the first condition of Reconstruct never fires.

Part (c). If $S = \{x : \sigma_{\min}(D\Phi_h^{(L)}|_x) = 0\}$ has $\mu(S) > 0$, then for $x \in S$ there exist sequences $x'_k \to x$ with $|\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)|/d_X(x'_k, x) \to 0$. Any consistent estimator satisfies $\widehat{d}_L(x, x'_k) \leq |\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)| + 2\varepsilon_n \to 0$. The set of pairs where $\widehat{d}_L \leq \eta_n$ thus has positive $\mu \otimes \mu$-measure, and the kernel condition fails. $\square$

Remark 7.4 (The Markov bridge is honest)**.** The conjunction theorem's proof identifies the honest bridge: $\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$ (an algebraic identity) makes algebra separation and metric separation literally the same event (Lemma 7.1), not merely related. Uniform separation (US) ensures that this identity is *empirically accessible*: the estimated delay vectors $\hat{\Gamma}_h^{(n)}$ faithfully represent the true $\Phi_h^{(L)}$, so the observer sees what the theory guarantees. Without (US), the bridge lies: the estimated $\widehat{d_L}$ can diverge from d_L on a positive-measure set, making the two witnesses certify different objects.

8 Discussion

8.1 What is open

Polynomial mixing. The Algebra Theorem (Theorem 5.9) and the Elbow Location Theorem (Theorem 5.7) require exponential mixing with resolvability constant $C_\lambda > 4$. Under polynomial mixing $\delta(L) \lesssim L^{-\alpha}$, the elbow drop at $L^*(n)$ satisfies $\Delta(L^*(n)) \sim \varepsilon_n / L^*(n) \to 0$ relative to ε_n: the elbow and the noise floor merge asymptotically and cannot be separated. A different stopping criterion (e.g. integrated elbow, or a bandwidth-adaptive rule) is needed. This is left open.

The $d > 2s$ hypothesis. Proposition 4.3 requires $d > 2s$ for the Rademacher concentration to be useful at rate ε_n. When $d \le 2s$, localised Rademacher complexity (Bartlett–Bousquet–Mendelson [1]) gives adaptive concentration without this condition but requires additional work. This is identified but not proved here.

Uniform separation from first principles. Condition (US) is a hypothesis in Theorems 6.6 and 7.3. The natural conjecture is that T Anosov with h generic implies (US) via the uniform hyperbolicity constants. This would close the gap between (G1)–(G2) and (US) in the hyperbolic category.

Stochastic T. For stochastic dynamics, $\Pi_h^{(L)}(x, \cdot)$ is not a Dirac delta. The TV separation d_L is no longer binary, and Hölder continuity of $x \mapsto \Pi_h^{(L)}(x, \cdot)$ in TV is both necessary and non-trivial. The four-link chain of Proposition 4.7 adapts, but the Conjunction Theorem's proof structure changes substantially.

Sharp rates for the data-driven lag. Theorem 5.7 shows $P(\hat{L}^* = L^*(n)) \to 1$ but does not bound the fluctuations of $\hat{L}^*$ around $L^*(n)$ on the event $\{\hat{L}^* \neq L^*(n)\}$. A Berry–Esseen-type result for $\hat{L}^* - L^*(n)$ would sharpen the stopping rule.

Concentration of the entropy witness. The empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ is a computationally simpler stopping criterion (Remark 5.13). Asymptotic equivalence with $\hat{\delta}(L, n)$ is established under fibre mixing (Corollary 5.12), but a finite-n concentration result for $\hat{H}_2(\nu_L^{(n)})$ — comparable to Proposition 4.3 for $\hat{\delta}(L, n)$ — is not proved here. Such a result requires only McDiarmid-type bounds on the U-statistic $\frac{1}{n^2} \sum_{i,j} \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]$, with no new structural assumptions beyond those of this paper. This is the most immediate extension of the present work.

8.2 What is settled

The four-paper programme is now complete at the level of:

- Measure-theoretic foundations (Paper I).
- Koopman–Perron duality and prediction (Paper II).
- Reconstruction theorem and the density bridge (Paper III).
- Finite-sample rates, witnesses, and the honest bridge (Paper IV).

The honest bridge theorem says: algebra reconstruction and kernel separation are the same event (Lemma 7.1), both are certifiable from data (Theorems 5.9 and 6.6), and their conjunction is provably reliable when reconstruction and (US) hold and provably unreliable when either fails (Theorem 7.3). This is not a claim that the two witnesses are informative about everything. It is a claim that they are informative about each other, and that the Markov structure of the delay vector — $\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$ — is why.

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